Chemistry 3A – Spring 2026 **Version A**

Sections 19320-19321

Midterm Examination #4

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt.
TEAR OFF LAST PAGE and use it as scratch for writing calculations.

LS11 means Lectures Set 11, with slide number ranges indicating where answer can be found

1. In a thermochemical equation, the energy value written on the reactant side indicates:
   a) The reaction is endothermic b) The reaction requires a catalyst
   c) The reaction is exothermic d) The reaction occurs at constant volume.

LS13, slides 50-52

2. Which statement accurately defines the **limiting reactant** in a chemical reaction?
   a) The reactant with the largest molar mass used in the reaction
   b) The reactant remaining in the largest amount after the reaction is complete
   c) The reactant with the smallest coefficient in the balanced chemical equation
   d) The reactant that determines the maximum amount of product that can be formed

LS11, slide 19

3. At a certain temperature, the reaction H_2 (g) + Cl_2 (g) ⇌ 2 HCl (g) reaches equilibrium and concentrations are determined: [H_2] = 0.0001 M, [Cl_2] = 0.0001 M, and [HCl] = 10,000 M. Compute the value of equilibrium constant *K*?
   a) 1×10^{12} b) 1×10^{-12} c) 1×10^{-16} **d) 1×10^{16}**

LS13, slides 29-30

$$K = \frac{[\mathrm{HCl}]^2}{[\mathrm{H_2}][\mathrm{Cl_2}]} = \frac{(10000)^2}{(0.0001)(0.0001)} = 1.0 \times 10^{16}$$

4. Which of the following is a FALSE statement about acid-base reactions?
   a) Titration is a quantitative technique that can determine the amount of moles of acid reacting with a base
   b) The products of the neutralization reaction are water (H_2O) and a salt
   c) There is a movement of a proton (H+) from the acid to the base
   d) Red litmus paper turning blue indicates a base, not an acid

All of the statements are basically TRUE, and none is false. This question will be marked correct for everyone.
The intention was to prepare the question as follows:

> Which of the following is a FALSE statement about acid-base reactions?
> a) Titration is a quantitative technique that can determine the amount of moles of acid reacting with a base
> b) The products of the neutralization reaction are water (H_2O) and a salt
> c) There is a movement of a proton (H+) from the base to the acid
> d) Red litmus paper turning blue indicates a base, not an acid

Which choice would you make then?

5. Which of the following pairs are conjugates in acid-base chemistry? (hint: weak acid or weak base)

a) HCl / Cl^- b) NH_4^+ / CH_4 **c) H_2SO4 / $HSO4^-$** **d) CH3COOH / $CH3COO^-$**

Here again is another mistake in the question that actually made it possible to select any one of three choices and have it be correct. Choices (a), (c), and (d) are actually conjugate pairs of an acid and their conjugate base. That is, they have a "HA / A^-" pattern, where to the left of the slash is the acid form, and when it loses a proton (H^+), it forms the conjugate base, which happens to be a -1 ion (it is not always required to be a -1 ion: it is only about losing a proton with its +1 charge and the resulting compound that remains after losing that proton).

The question should be "Which of the following pairs are conjugates **useful in buffers** in acid-base chemistry? (hint: weak acid or weak base)". The words added in bold.
When "useful in buffers" is added, it was stated that weak acids and weak bases were actually useful as buffers. Strong acids and bases are not useful as buffers. Choice (a) with the "HCl" acid would not be a good buffer since HCl is a strong acid. Neither would choice (c) be a good choices, since H_2SO_4 (sulfuric acid) is also a strong acid. The only correct answer would be choice (d), CH_3COOH / CH_3COO^- (acetic acid/acetate). It has been indicated that organic acids and bases (carbon-based compounds) have been indicated as being weak acids and bases, and thus ideal compounds for buffers (compounds that resist change in pH when strong acids and bases are mixed with them).

What can be understood from choice (b)? The use of "CH_4" in the choice made that pairing nonsense. But suppose choice (b) was instead "NH_4^+ / NH_3". Well that now does indicate a true acid/conjugate base pairing, because NH_4^+ can be a proton donator, and when it does, it forms NH_3, the conjugate base. Even more interesting is that NH_3 is a weak base (!) and its conjugate acid NH_4^+ is thus a weak acid, and this pairing joins CH_3COOH / CH_3COO– (acetic acid/acetate) as a buffer!

LS12, slides 24-29

Students who selected choice (a), (c), and (d) were scored as correct

6. Which statement is TRUE about pure water (H_2O)?

a) It has an equilibrium constant K_w whose value = 10^{-7}

b) It has a pH = 7.0

c) It has a $[H^+] = 10^{-14}$ *M*. ([H+] is also $[H_3O^+]$)

d) It has the ability to act as an acid, but not as a base

LS11, slide 19. The equilibrium constant $K_w = 10^{-14}$, so (a) cannot be true. As for choice (c), the $[H^+] = 10^{-7}$ M, and so is $[OH^-] = 10^{-7}$ M. In fact $K_w = [H^+][OH^-] = 10^{-14}$. If you convert $[H^+] = 10^{-14}$ *M* into a pH using the pH definition, it becomes a pH = 14, which contradicts what is known about the pH of pure water, which is pH = 7. Choice (d) is incorrect since water has been stated as acting as an acid and base.

7. For the balanced reaction : CH_4 (g) + 4 Cl_2 (g) → CCl_4 (l) + 4 HCl (g), you start with 100.0 g methane (CH_4). How much **carbon tetrachloride** (CCl_4) will you produce? The molar masses (g/mol) are given: CH_4: 16.04, Cl_2: 70.91, CCl_4: 153.81, HCl: 36.46. (Hint: you should only need two of the four)

a) 239.7 g b) 10.43 g **c) 958.9 g** d) 153.8 g

LS11, slide 19. Your knowledge of stoichiometry is challenged here. You are given the fact that one of the reactants, CH4 (methane) is present in 100.0 g amounts. You are then asked what how much CCl4 (carbon tetrachloride) can theoretically be produced. You need to immediately think of a "reaction ratio", and you see in the balanced equation that 1 mol CCl4 can be produced for every 1 mol CH4. Set up an equation now. To the right of the equal sign, you have "? mol CCl_4". To the left of the equal sign is everything else you need that makes that calculation.

Everything on either side of the reaction arrow must first be converted to moles. If you have a mass of compound, one needs to convert the mass in grams to moles, using the molar mass. If you have a volume of a solution, you need to know the concentration of the compound (in molar, or mol/L) which can convert to moles. If you have a gas amount, you need to use the ideal gas law with known parameters of pressure (*P*), volume (*V*), and temperature (*T*), along with the gas constant *R*, to get to moles (*n*). All these calculations have to get to moles on both sides of the reaction arrow, and then additional "back" calculations might be made if you are asked for an answer in "grams compound", or "liters solution" or pressure or volume or temperature of a gas.

Start out by putting "100.0 g CH4" on the left side of the equal sign:

$$100.0\ g\ CH_4\ \times \ldots = ?\ g\ CCl_4$$

Note the question asked "How much CCl4 will you produce?". Units were not given in the question: is it moles, which is an amount? Is it mass in grams? You can see from the choices it asks for grams, so it is a mass.

In analyzing how to get to the answer, remember that chemical calculations are set up to get quantities (numbers with units) to cancel, which units in the numerator canceling units in the denominator. We have to cancel "g CH4": how is it done? Using the molar mass of CH4 is the path forward. Try it.

$$100.0\ \text{g}\ \text{CH}_4\ \times \frac{1\ \text{mol}\ \text{CH}_4}{16.04\ \text{g}\ \text{CH}_4} \times \ldots = ?\ \text{g}\ \text{CCl}_4$$

Now seek to cancel "mol CH_4": that would be with the reaction ratio of CH_4 and CCl_4

$$100.0\ \text{g}\ \text{CH}_4\ \times \frac{1\ \text{mol}\ \text{CH}_4}{16.04\ \text{g}\ \text{CH}_4} \times \frac{1\ \text{mol}\ \text{CCl}_4}{1\ \text{mol}\ \text{CH}_4} \times \ldots = ?\ \text{g}\ \text{CCl}_4$$

The left side of the equal sign will leave a result of "mol CCl4" and the wanted result is "g CCl4", so another factor is necessary. That would the molar mass CCl4.

$$100.0\ \text{g}\ \text{CH}_4\ \times \frac{1\ \text{mol}\ \text{CH}_4}{16.04\ \text{g}\ \text{CH}_4} \times \frac{1\ \text{mol}\ \text{CCl}_4}{1\ \text{mol}\ \text{CH}_4} \times \frac{153.81\ \text{g}\ \text{CCl}_4}{1\ \text{mol}\ \text{CCl}_4} = \ 958.9\ \text{g}\ \text{CCl}_4$$

The result is shown because all units cancel and what is left are the units which are the target units. Dimensional analysis is the process by which chemical calculations are solved.

8. Which of the following statements is TRUE about atomic fission and atomic fusion

a) The sun generates energy through atomic fission reactions

b) Nuclear power plants generate energy through atomic fission reactions

c) Nuclear power plants generate energy through atomic fusion reactions

d) The use of enriched uranium-235 isotope is a good example of a fusion chain reaction

LS11, slide 19. Choice (a) can't be true because the Sun was shown to produce its energy through the ***fusion*** of basic hydrogen-1 isotopes. Choice (c) is not true because, while there is research to get energy/power from fusion reactions (based on fusion of hydrogen-2 and hydrogen-3 isotpes), it has not yet been obtained. Choice (d) talks about Uranium-235 which is radioactive and used for nuclear power (and also weapons), it releases this energy through a ***fission*** chain reaction and not ***fusion***, so there is no concept of a fusion chain reaction here**.** Choice (b) is the correct one, with U-235 being struck by neutrons, and this reaction creates the instability necessary to initiate fission. One of the main points of Lecture Set 14 in the course is to understand what fission and fusion are and the key distinctions of the atom and their nuclei. This examination did not test on the quantitative objectives of nuclear chemistry and only on qualitative objectives, considered to be easier to understand.

9. Of the statements below, select the best that is true of $^{31}_{15}\mathrm{P}$?

a) It has 46 neutrons and protons

b) It has 31 protons

c) It has 16 neutrons

d) It has 15 neutrons

LS11, slide 19. This also goes back to understanding the superscript to the left of the symbol being the mass number (A), equal to protons plus neutrons, and the subscript to the left of the symbol being atomic number (Z), equal to the protons. Choice (a) is false: the sum of the A and Z numbers is meaningless. Choice (b) is false: there are 15 protons, the Z value. Choice (d) is false: the A number subtracted by the Z number (31 – 15) is the number of neutrons: 16, thus choice (c)

10. A solution has $[H^+] = 1.0 \times 10^{-9}$ M. What is the pH?

a) 1.00 b) 5.00 **c) 9.00** d) 14.00

LS11, slide 19. This tests knowledge of converting pH and $[H^+]$: pH = -log$[H^+]$. Thus pH = -log(1.0 x 10–9). Plug it in the calculator: pH = 9.00.

11. Which of the following is a strong acid?

a) KOH **b) HCl** c) NaCl d) CO_2

LS11, slide 19. KOH is a strong base. NaCl and CO2 are neither acids or bases, weak or strong. HCl has been identified as a strong acid.

12. 100.00 mL of acetic acid solution was titrated with 0.100 M NaOH solution with the initial buret reading at 3.00 mL and the final buret reading at 18.00 mL. What is the acetic acid concentration?

a) 0.00150 M **b) 0.0150 M** c) 0.0300 M d) 0.150 M

LS11, slide 19. Ultimately, this is about a titration in which an acid or base neutralizes the other. This X moles acid being equal to X moles base for the neutralization. This straight from Experiment 12b, which you must understand at a basic level for introductory chemistry. It is important to know this titration method is a $C_1V_1 = C_2V_2R$ problem, where C1 = NaOH titrant, V1 = NaOH volume, V2 = volume of HCl. The value R the reaction ratio between the acid and base. Acetic acid (CH_3COOH) has one H+ reacting with one OH from NaOH. In Experiment 12b, citric acid was tested, and its structure has three H+ reacting with one OH from NaOH, so R would 3 NaOH / 1 citric acid. In another part, oxalic acid was titrated, and it has two H+ reacting with NaOH titrant, so it was 2 NaOH / 1 oxalic acid. Suppose that C2 is acetic acid concentration, which is the unknown. Let's set it up:

$$C_1V_1 = C_2V_2R$$

Do algebra to get C_2 on one side of equal sign, then substitute values:

$$C_2 = \frac{C_1V_1}{V_2R} = \frac{(0.100\ \text{mol/L NaOH})[(18.00\ \text{mL} - 3.00\ \text{mL})\ \text{NaOH}]}{(100.00\ \text{mL}\ CH_3COOH) \times \frac{1\ \text{mol NaOH}}{1\ \text{mol}\ CH_3COOH}}$$

It's important to understand that the units will cancel in a way that leaves mol/L (M). To compute the volume of NaOH used, it was necessary to subtract from the final buret reading the initial, with a 15.00 mL volume. The simplified expression becomes (0.100 M)(15.00 mL)/(100.00 mL) = 0.0150 M with three significant digits (limited by the "0.100 M" value for NaOH concentration)

13. If a weak acid has an equilibrium constant $K_a = 5.0 \times 10^{-8}$, then what is the equilibrium constant K_b of its conjugate base? Note $K_w = 1.0 \times 10^{-14}$

a) 2.0×10^{-6} **b) 2.0×10^{-7}** c) 5.0×10^{-7} d) 5.0×10^{-22}

LS13, slide 64. Important in chemistry of acids and bases is the concept that the equilibrium constant for an acid multiplied by the equilibrium constant for a base equals the equilibrium constant for water, that is, $K_w = K_aK_b$. The question asks for K_b given K_a and K_w. Use the algebra to solve for K_b ($K_b = K_w / K_a$), then substitute values: $K_b = (1.0 \times 10^{-14}) / (5.0 \times 10^{-8})$. Just enter the expression into the calculator now to get the result: $K_b = (1.0 \times 10^{-14}) / (5.0 \times 10^{-8}) = 2.0 \times 10^{-7}$

14. Consider the following equilibrium: $PCl_3\ (g) + Cl_2\ (g) \rightleftharpoons PCl_5\ (g)$. If large amount of PCl_3 reactant is added to container, in which direction will the reaction shift to restore equilibrium?

a) Shift to the right (forward reaction) to produce more PCl_5.

b) Shift to the left (reverse reaction) to consume the added PCl_3.

c) There will be no shift, as adding PCl_2 reactant does not affect the equilibrium

d) The equilibrium constant K will decrease

LS13, slides 46-49. This is about the "concentration stress" concept of Le Chatelier's Principle. If more reactant is added, then this favors a shift to the making of product to restore equilibrium. If more of any compounds on the product side (right of equilibrium symbol) were added, the shift would be to the left of the symbol favor formation of reactants in restoring equilibrium.

15. Which type of radiation has the most ***ionizing*** power?

a) Alpha particles
b) Beta particles
c) Gamma rays
d) None of the above

LS14 slides 6 and 25

16. What is the radioisotope of an element that is a gas that can accumulate in homes and is responsible particularly for causing nonsmoking-related lung cancers?

a) Xenon-133 b) Krypton-85 **c) Radon-222** d) Carbon-14

LS14, slide 26

17. X, Y, and Z are gaseous compounds in the reaction X (g) + Y (g) ⇌ **2** Z (g). What will happen if the pressure is **increased** on the system according to Le Chatelier's Principle?

a) The reaction will favor the formation of the reactants (the reverse direction)
b) The reaction will favor the formation of the products (the forward direction)
c) The reaction will change the reaction to **2** X (g) + Y (g) ⇌ Z (g)
d) There will be no effect: the amounts of reactants and product remain the same

LS13, slides 53-54. This is a "pressure stress" concept related to Le Chatelier's Priniciple. If the total pressure on a system filled with gases is increased, the shift to restore equilibrium will be to the side of the equilibrium symbol with the fewest moles of all gases; if pressure is reduced, it will favor the side with the greater number of moles of all gases. Let's look at the reaction: on the left, the X (g) + Y (g) has two moles of all gases (1 mole of X, 1 mole of Y), and on the right, there are two moles of all gases (2 moles Z). So the moles of gas on the left and right of equilibrium are equal! When the moles are equal, there is no "pressure stress" effect, and so increasing or decreasing system pressure will cause no shift in the forward or reverse direction of the reaction. This makes choices (a) and (b) not applicable in this case. Choice (c) is about changing the nature of the reaction which has never been discussed or stated (this choice is a distractor).

18. Initially you have 800.0 g of U-235 radioisotope. How much do you have left if four (4) half-lives pass?

a) 800.0 g b) 400.0 g c) 100.0 g **d) 50.00 g**

LS14, slide 13-15. The concept of a "half-life" is not just important to understanding radioactivity and its decay, but it is seen in many of the STEM disciplines. So understanding it is not just important to chemistry, but many other sciences. If one half life passes, that means one half of the substance in question is left. The math is simple: Amount left = original amount × $(1/2)^n$, where n = the number of half lives. If one half life passes, then n = 1, and the amount left = original amount × $(1/2)^1$ which basically one-half. So if four half lives pass, then amount left = original amount × (1/2)4 = original amount × (1/16). Thus 800.0 g × (1/16) = 50.00 g. You could also just halve 800.0 g four times: 400.0, 200.0, 100.0, 50.00, another way of doing the math.

19. One of the statements below is FALSE about what a buffer is and its properties?

a) The best buffers are weak acids or bases their conjugate pairs
b) it can resist changes in solution pH caused by addition of strong acids or bases
c) **It is a term that applies to shielding for all radiation, alpha and beta particles and gamma rays**
d) Formic acid and its conjugate base formate were mentioned as examples of buffers

LS13, slides 42-45. Choices (a), (b) and (d) were mentioned in that slide range. Choice (c) is so false and never mentioned in the lecture that it seems the question is a gift with just minimal studying.

20. For the equilibrium $Ca(OH)_2$ (s) ⇌ Ca^{2+} (aq) + 2 OH^- (aq), K_{sp} = 5.0 x 10^{-6}. (The units are not given usually, but value is K_{sp} = 5.0 x 10-6 M^3.) If $[Ca^{2+}]$ = 1.0 × 10^{-1} M, what is $[OH^-]$?

a) 1.0 × 10^{-1} *M* b) 5.2 × 10^{-5} *M* **c) 7.1 × 10^{-3} M** d) 2.5 × 10^{-5} *M*

LS13, slides 66-67. This is a solubility product calculation. You first have to write the solubility product as the multiplication of all aqueous phase compounds AND you have make sure the coefficients are made into exponents in the K_{sp} math expression, just like for any equilibrium constant. The solublility product chemical equation is shown as the solid phase reactant (left side of equilibrium symbol) and all aqueous phase compounds to the right of the symbol. Now write the Ksp: $K_{sp} = [Ca^{2+}] \times [OH^-]^2$. (it is unnecessary to write the multiplication symbol, but it is shown to illustrate it. Now substitute the values and the constant: 5.0 x 10^{-6} = (1.0 x 10^{-1}) × $[OH^-]^2$. The rest is math/algebra*:* $[OH^-] = \sqrt{\frac{5.0 \text{ x } 10^{-6}}{1.0 \text{ x } 10^{-1}}} = 7.1 \text{ x } 10^{-3}$ *M*

21. Which one of the choices below is a TRUE statement about a catalyst?

a) It *increases* the activation energy (E_a) seen in the reaction coordinate diagram

b) it is *never* consumed or produced in the chemical reaction like reactants or products

c) It *stops* a chemical reaction immediately

d) it affects the enthalpy (ΔH) of the reaction

LS13, slide 13. Choice (a) is the opposite of what a catalyst does: it DECREASES the activation energy (the height of the "hump" in the reaction progress plot). Choice (c) was never mentioned: false. Choice (d) is false as seen in the reaction progress plot as ΔH is never changed.

22. What is TRUE about the term ***molarity*** and its use?

a) The units of molarity are "mol/L" (moles per liter)

b) it is used to indicate the concentration of a solution

c) if you multiply it by a volume in liters, you can get the amount (moles) of a substance

d) all of the above

LS11. This also goes back to LS9 as well. These are fundamentals of concentration and the concept of molarity. Units can be mol/L which is also symbolized by M. And if you multiply by the volume, you get the actual amount of the compound in moles. Choices (a), (b) and (c) are all true.

23. What is FALSE about the concept of chemical equilibrium?

a) it indicates chemical reactions can go in a forward and reverse direction

b) when equilibrium is attained, the rate of the forward and reverse reactions are equal

c) it only applies to what happens when the reactants and products are gases

d) equilibrium can be quantitatively understood by a constant *K*

LS13, slides 13, 17, 21. Choice (a) is a fundamental of chemical equilibrium. Choice (b) is also noted as a concept. Choice (d) is about the quantitative understanding of equilibrium. Choice (c) is false because equilibrium applies to compounds in the gas phase, certainly the aqueous phase.

24. According to the Bronsted-Lowry definition of acids and bases, which statement is TRUE?

a) an acid is a proton acceptor

b) an acid is a proton donor

c) a base is a proton donor

d) none of the above

LS12, slide 17

25. Which of the following factors is LEAST LIKELY IF AT ALL to affect the rate of a chemical reaction?
  a) temperature
  b) concentration
  c) orientation of collisions
  d) the geometry of the container where the reaction is done

LS13, slides 3, 7-10. Choice (a), (b) and (c) are all important and mentioned as affecting the rate of a chemical reaction. The lectures never mentioned choice (d) as a factor. That is not to say that perhaps a reaction vessel's geometry could not affect reaction rates, but that was not mentioned and any effect would be minimal to none compared to the other factors.

TRUE OR FALSE

26. A catalyst will reduce the enthalpy Δ*H* of either an endothermic or exothermic reaction
  a) true **b) false**

In a multiple-choice question, the effect of a catalyst on ΔH was said not to happen.

27. If the temperature of an endothermic reaction at equilibrium is increased, the equilibrium will shift in the endothermic direction.
  a) true b) false

LS13, slides 50-51

28. **Gamma rays** are the most penetrating type of common radiation and require thick concrete or lead shielding
  a) true b) false

LS14, slides 5, 6, 25

29. Strong acids and like HCl and strong bases like NaOH are called that because in aqueous solutions, they exist completely as ions and not as HCl (aq) or NaOH (aq)
  a) true b) false

LS12, slide 5

30. Balancing a chemical reaction equation is done by a process of adjusting the coefficients of all of the compounds in the chemical reaction equation.
  a) true b) false

LS12, slide 5

31. The half-life of a radioisotope is the time required for half of the sample to decay.
  a) true b) false

LS12, slide 5

32. In performing a chemical reaction, the percent yield is the theoretical yield divided by the actual yield times 100 percent (100%)
  a) true **b) false**

LS11, slide 26. It's the actual yield divided by the theoretical yield, not the way it was worded.

33. Collision theory in part states that for a chemical reaction to occur, particles must collide with correct orientation and sufficient energy
  a) true b) false

LS13, slide 3

34. Water (H_2O) can act as both an acid and a base depending on the reaction partner

a) true b) false

LS13, slide 64

35. The sum of the pH and the pOH is always 14.

a) true b) false

LS13, slide 34